Lecture Notes on Limits, Continuity, and Differentiation

A first rigorous course in mathematical analysis

Prepared as lecture notes

May 16, 2026

These notes are written for a mathematics lecture. The goal is not only to remember formulas, but to understand the logical chain:

$$\text{limit} \implies \text{continuity} \implies \text{differentiability} \implies \text{local linear approximation}.$$

The emphasis is on definitions, geometric intuition, proofs, standard examples, and the difference between formal and computational reasoning.

Contents

1 Why these topics belong together **4**

2 Functions, graphs, and neighborhoods **4**

2.1 Functions as assignments 4

2.2 Intervals and punctured neighborhoods 4

2.3 Graphical intuition 5

3 Limits of functions **5**

3.1 Informal definition 5

3.2 The epsilon-delta definition 6

3.3 First epsilon-delta examples 6

3.4 One-sided limits 7

3.5 Algebra of limits 8

3.6 Indeterminate forms 8

3.7 Factoring and cancellation 9

3.8 Squeeze theorem . . . 9
3.9 Infinite limits and vertical asymptotes . . . 9
3.10 Limits at infinity . . . 10

4 Important standard limits **11**
4.1 Trigonometric standard limits . . . 11
4.2 Exponential and logarithmic standard limits . . . 11

5 Continuity **11**
5.1 Definition of continuity at a point . . . 11
5.2 Three conditions for continuity . . . 12
5.3 Types of discontinuity . . . 12
5.4 Continuity on intervals . . . 12
5.5 Algebra of continuous functions . . . 13
5.6 Composition of continuous functions . . . 13

6 The intermediate value theorem and consequences **13**
6.1 The intermediate value theorem . . . 13
6.2 Existence of roots . . . 14
6.3 Extreme value theorem . . . 14

7 Differentiation **14**
7.1 Average rate of change . . . 14
7.2 Derivative as a limit . . . 15
7.3 Equivalent form . . . 15
7.4 Derivative of elementary functions from the definition . . . 15
7.5 Differentiability implies continuity . . . 16
7.6 Tangent line and local linear approximation . . . 17

8 Rules of differentiation **17**
8.1 Linearity . . . 17
8.2 Product rule . . . 17
8.3 Quotient rule . . . 18
8.4 Chain rule . . . 18
8.5 Power rule . . . 18

9 Derivatives of standard functions **19**

10 Higher derivatives and shape of a graph **19**

10.1 Second derivative 19

10.2 Monotonicity 19

10.3 Concavity 20

10.4 Critical points 20

11 Rolle's theorem and the mean value theorem **20**

11.1 Rolle's theorem 20

11.2 Mean value theorem 21

11.3 Consequences of the mean value theorem 21

12 Taylor's first approximation **21**

13 Non-differentiability: what can go wrong? **22**

13.1 Corners 22

13.2 Cusps 22

13.3 Vertical tangents 22

13.4 Discontinuities 22

14 Worked examples **23**

15 Proof templates **24**

15.1 Template for proving a finite limit 24

15.2 Template for proving continuity 25

15.3 Template for proving differentiability 25

16 Exercises **25**

17 Selected solutions **26**

18 Compact formula sheet **28**

19 Final conceptual map **28**

1 Why these topics belong together

Limits, continuity, and differentiation form the first conceptual block of differential calculus. Their order is not accidental.

- A **limit** describes what values of a function approach near a point, even if the function is not defined at that point.
- **Continuity** says that the value at the point agrees with the limiting behavior near the point.
- A **derivative** is a limit of slopes. It measures the best local linear approximation to a function.

The central idea is that many mathematical statements about local behavior are not statements about one point alone. They are statements about what happens in arbitrarily small neighborhoods of that point.

Local point of view

A property is called *local near* a if it depends only on the behavior of the function on some interval around a, possibly excluding the point a itself.

This distinction is essential. A limit at a may exist even if the function is undefined at a. Continuity at a cannot be discussed unless the function is defined at a. Differentiability at a is even stronger: it requires the function to be well approximated by a line near a.

2 Functions, graphs, and neighborhoods

2.1 Functions as assignments

A real function is a rule

$$f : D \to \mathbb{R},$$

where $D \subseteq \mathbb{R}$ is the domain. For each $x \in D$, the function assigns exactly one number $f(x)$.

In analysis, the domain matters. The functions

$$f(x) = x^2, \qquad x \in \mathbb{R},$$

and

$$g(x) = x^2, \qquad x \geq 0,$$

have the same formula but different domains, and therefore different local behavior at boundary points.

2.2 Intervals and punctured neighborhoods

For $a \in \mathbb{R}$ and $\delta > 0$, the interval

$$(a - \delta, a + \delta)$$

is called a neighborhood of a. A punctured neighborhood is the same interval with the point a removed:

$$0 < |x - a| < \delta.$$

This notation means:

$$x \in (a - \delta, a + \delta) \quad \text{and} \quad x \neq a.$$

Why punctured neighborhoods appear in limits

When studying $\lim_{x \to a} f(x)$, the value $f(a)$ is irrelevant. A limit is about values of $f(x)$ for x close to a, not necessarily equal to a.

2.3 Graphical intuition

A graph is often the first source of intuition, but it is not a proof. A graph suggests what might be true; definitions and theorems explain why it is true.

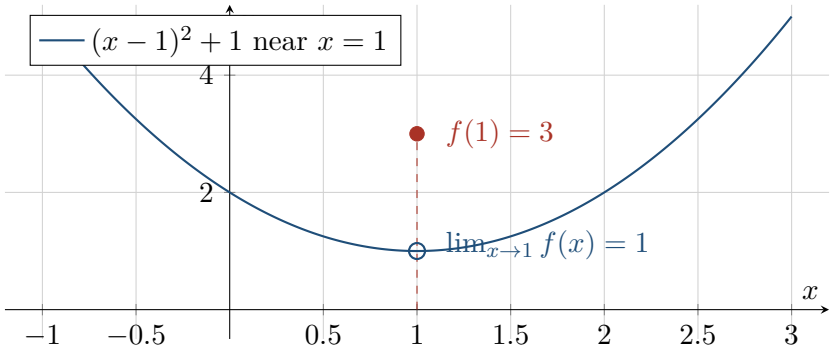


Figure 1: A limit may exist even when the value of the function at the point is different from the limiting value.

The figure illustrates one of the main lessons of limits: the value at the point is not what determines the limit.

3 Limits of functions

3.1 Informal definition

The statement

$$\lim_{x \to a} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a, but not necessarily equal to a.

The key words are *arbitrarily close*. They indicate that a limit is not about one fixed level of approximation. It must work for every required accuracy.

3.2 The epsilon-delta definition

Limit of a function at a point

Let $f : D \to \mathbb{R}$, let a be an accumulation point of D, and let $L \in \mathbb{R}$. We say that

$$\lim_{x \to a} f(x) = L$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in D$,

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

This definition is the formal language of local approximation.

- ε measures the demanded accuracy in the output.
- δ measures how close the input must be to a.
- The phrase *for every* $\varepsilon > 0$ means the approximation must be possible at every scale.
- The phrase *there exists* $\delta > 0$ means the allowed input range may depend on ε.

Reading an epsilon-delta proof

An epsilon-delta proof usually has two parts.

1. Scratch work: find a condition on δ that would make $|f(x) - L| < \varepsilon$ true.
2. Formal proof: given $\varepsilon > 0$, choose such a δ, then prove the implication.

3.3 First epsilon-delta examples

Example: proving a linear limit

Prove that

$$\lim_{x \to 2} (3x - 1) = 5.$$

We need to make

$$|(3x - 1) - 5| = |3x - 6| = 3\,|x - 2|$$

smaller than ε. It is enough to require

$$|x - 2| < \frac{\varepsilon}{3}.$$

Thus, given $\varepsilon > 0$, choose

$$\delta = \frac{\varepsilon}{3}.$$

Then whenever $0 < |x - 2| < \delta$, we have

$$|(3x - 1) - 5| = 3\,|x - 2| < 3\delta = \varepsilon.$$

Therefore the limit is 5.

Example: proving a quadratic limit

Prove that

$$\lim_{x\to 1} x^2 = 1.$$

We estimate

$$\left|x^2 - 1\right| = |x-1|\,|x+1|\,.$$

The factor $|x-1|$ can be made small directly. The factor $|x+1|$ must be controlled near $x = 1$.
Assume first that $|x-1| < 1$. Then $0 < x < 2$, so $|x+1| < 3$. Hence

$$\left|x^2 - 1\right| < 3\,|x-1|\,.$$

Given $\varepsilon > 0$, choose

$$\delta = \min\left\{1, \frac{\varepsilon}{3}\right\}.$$

If $0 < |x-1| < \delta$, then

$$\left|x^2 - 1\right| < 3\,|x-1| < 3\delta \le \varepsilon.$$

Therefore $\lim_{x\to 1} x^2 = 1$.

A common mistake

In a proof, one cannot simply set $\delta = \varepsilon$ unless that choice has been justified. The choice of δ must be strong enough to force the desired inequality in the output.

3.4 One-sided limits

Sometimes the behavior from the left and from the right is different.

One-sided limits

We write

$$\lim_{x\to a^-} f(x) = L$$

if $f(x) \to L$ as x approaches a from the left, meaning $x < a$. Similarly,

$$\lim_{x\to a^+} f(x) = L$$

if $f(x) \to L$ as x approaches a from the right, meaning $x > a$.

Two-sided limit criterion

The two-sided limit $\lim_{x\to a} f(x) = L$ exists if and only if both one-sided limits exist and are equal:

$$\lim_{x\to a^-} f(x) = L \qquad \text{and} \qquad \lim_{x\to a^+} f(x) = L.$$

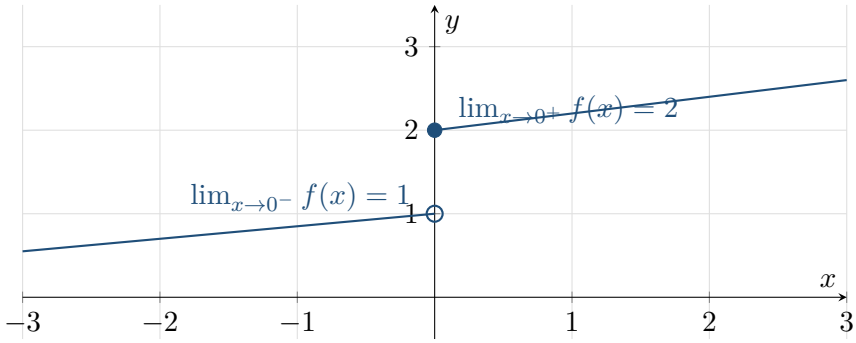


Figure 2: A jump discontinuity: the left and right limits are different, so the two-sided limit does not exist.

3.5 Algebra of limits

Limit laws

Assume that

$$\lim_{x\to a} f(x) = A, \qquad \lim_{x\to a} g(x) = B.$$

Then:

$$\begin{aligned}
\lim_{x\to a}(f(x)+g(x)) &= A+B,\\
\lim_{x\to a}(f(x)-g(x)) &= A-B,\\
\lim_{x\to a}(f(x)g(x)) &= AB,\\
\lim_{x\to a}\frac{f(x)}{g(x)} &= \frac{A}{B}, \qquad B\neq 0.
\end{aligned}$$

For every constant c,

$$\lim_{x\to a} cf(x) = cA.$$

These laws justify much of the computational practice of evaluating limits. However, they apply only when the component limits exist and the resulting expression is meaningful.

3.6 Indeterminate forms

Expressions such as

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad 0\cdot\infty, \quad \infty-\infty$$

are called indeterminate forms. They do not have a predetermined value. They indicate that further analysis is needed.

Example: same algebraic form, different limits

As $x\to 0$:

$$\frac{x}{x}\to 1, \qquad \frac{x^2}{x}\to 0, \qquad \frac{x}{x^2}\to\infty \text{ in magnitude.}$$

All three expressions are of the symbolic type $0/0$, but their limiting behavior is different.

3.7 Factoring and cancellation

Example: removable singularity

Compute

$$\lim_{x\to 2} \frac{x^2-4}{x-2}.$$

For $x \neq 2$,

$$\frac{x^2-4}{x-2} = \frac{(x-2)(x+2)}{x-2} = x+2.$$

Therefore

$$\lim_{x\to 2} \frac{x^2-4}{x-2} = \lim_{x\to 2}(x+2) = 4.$$

The original expression is undefined at $x = 2$, but the limit exists.

3.8 Squeeze theorem

Squeeze theorem

Suppose that for all x sufficiently close to a, except possibly at a,

$$g(x) \leq f(x) \leq h(x).$$

If

$$\lim_{x\to a} g(x) = \lim_{x\to a} h(x) = L,$$

then

$$\lim_{x\to a} f(x) = L.$$

Example: a highly oscillatory function

Show that

$$\lim_{x\to 0} x^2 \sin\left(\frac{1}{x}\right) = 0.$$

Because

$$-1 \leq \sin\left(\frac{1}{x}\right) \leq 1,$$

we get

$$-x^2 \leq x^2 \sin\left(\frac{1}{x}\right) \leq x^2.$$

Both bounding functions tend to 0. Hence the limit is 0.

3.9 Infinite limits and vertical asymptotes

The notation

$$\lim_{x\to a} f(x) = \infty$$

does not mean that ∞ is a real number. It means that $f(x)$ grows beyond every real bound as x approaches a.

Infinite limit

We write

$$\lim_{x \to a} f(x) = \infty$$

if for every $M > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad f(x) > M.$$

The definition of $\lim_{x \to a} f(x) = -\infty$ is analogous.

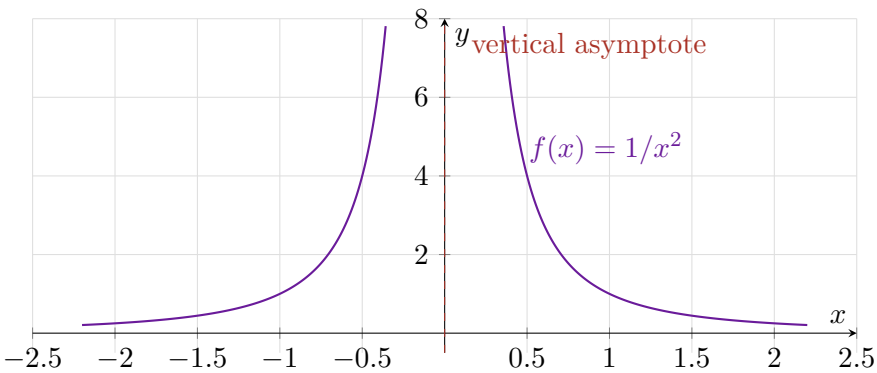


Figure 3: The function $1/x^2$ tends to $+\infty$ as $x \to 0$.

3.10 Limits at infinity

The expression

$$\lim_{x \to \infty} f(x) = L$$

means that $f(x)$ approaches L as x grows without bound.

Limit at infinity

We say that

$$\lim_{x \to \infty} f(x) = L$$

if for every $\varepsilon > 0$ there exists $R > 0$ such that

$$x > R \quad \Longrightarrow \quad |f(x) - L| < \varepsilon.$$

A horizontal asymptote is often a graphical representation of such a limit.

Example: rational function at infinity

Compute

$$\lim_{x \to \infty} \frac{3x^2 - 5x + 1}{2x^2 + x - 4}.$$

Divide numerator and denominator by x^2:

$$\frac{3 - 5/x + 1/x^2}{2 + 1/x - 4/x^2} \to \frac{3}{2}.$$

Therefore the horizontal asymptote is $y = 3/2$.

4 Important standard limits

Some limits occur so often that they become basic tools.

4.1 Trigonometric standard limits

Fundamental trigonometric limits

In radians,

$$\lim_{x\to 0}\frac{\sin x}{x}=1,\qquad \lim_{x\to 0}\frac{1-\cos x}{x}=0,\qquad \lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}.$$

The first of these is one of the most important limits in calculus. It is the reason why the derivative of $\sin x$ is $\cos x$ when angles are measured in radians.

4.2 Exponential and logarithmic standard limits

Exponential and logarithmic limits

For the natural exponential and logarithm,

$$\lim_{x\to 0}\frac{e^x-1}{x}=1,\qquad \lim_{x\to 0}\frac{\ln(1+x)}{x}=1.$$

Also,

$$\lim_{n\to\infty}\left(1+\frac{x}{n}\right)^n=e^x.$$

These limits connect calculus with growth processes, compound interest, differential equations, and many later topics.

5 Continuity

5.1 Definition of continuity at a point

Continuity at a point

Let $f : D \to \mathbb{R}$ and let $a \in D$. The function f is continuous at a if

$$\lim_{x\to a} f(x) = f(a).$$

Equivalently, for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x \in D$,

$$|x-a|<\delta \quad\Longrightarrow\quad |f(x)-f(a)|<\varepsilon.$$

The difference between the limit definition and continuity is subtle but important. In the limit definition we use $0 < |x-a| < \delta$, because $x = a$ is excluded. In continuity, the point $x = a$ is included.

5.2 Three conditions for continuity

A function is continuous at a precisely when:

1. $f(a)$ is defined;
2. $\lim_{x\to a} f(x)$ exists;
3. $\lim_{x\to a} f(x) = f(a)$.

If any of these fails, the function is not continuous at a.

5.3 Types of discontinuity

Type	Description
Removable	The limit exists, but the value is missing or different.
Jump	The left and right limits exist but are different.
Infinite	The function grows without bound near the point.
Oscillatory	The function oscillates too much for a limit to exist.

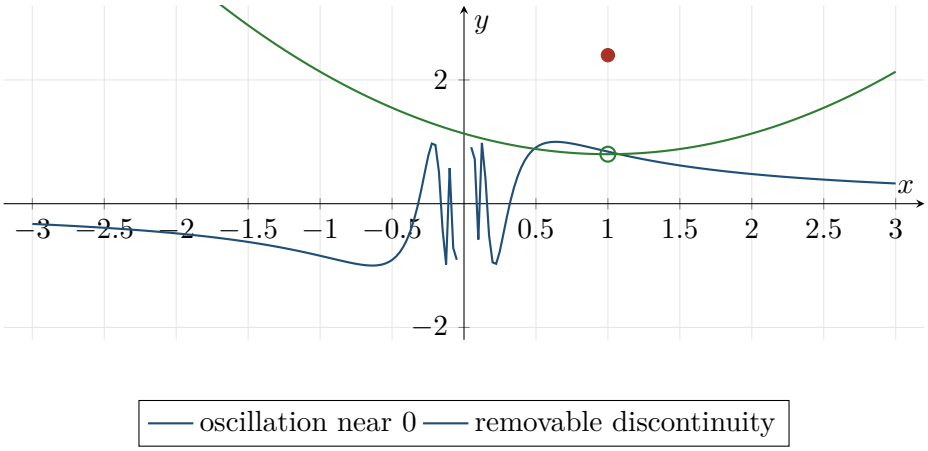


Figure 4: Two typical failures of continuity: oscillation near a point and a removable discontinuity.

5.4 Continuity on intervals

Continuity on an interval

A function is continuous on an interval I if it is continuous at every point of I. At endpoints, continuity is understood one-sidedly.

For example, a function continuous on $[a, b]$ is continuous from the right at a, from the left at b, and two-sidedly at all points of (a, b).

5.5 Algebra of continuous functions

Continuous functions are stable under algebraic operations

If f and g are continuous at a, then so are:

$$f + g, \quad f - g, \quad fg, \quad cf.$$

If also $g(a) \neq 0$, then f/g is continuous at a.

Polynomial functions are continuous everywhere. Rational functions are continuous at every point where the denominator is nonzero.

5.6 Composition of continuous functions

Continuity of composition

If g is continuous at a, and f is continuous at $g(a)$, then $f \circ g$ is continuous at a:

$$\lim_{x \to a} f(g(x)) = f(g(a)).$$

This theorem is one reason why we can evaluate many limits by substitution. For example,

$$\lim_{x \to 2} \sin(x^2) = \sin(4),$$

because x^2 and $\sin x$ are continuous.

6 The intermediate value theorem and consequences

6.1 The intermediate value theorem

Intermediate value theorem

Let f be continuous on $[a, b]$. If N lies between $f(a)$ and $f(b)$, then there exists $c \in [a, b]$ such that

$$f(c) = N.$$

Geometrically, a continuous graph cannot jump over a horizontal level. If it moves from below that level to above it, it must cross it.

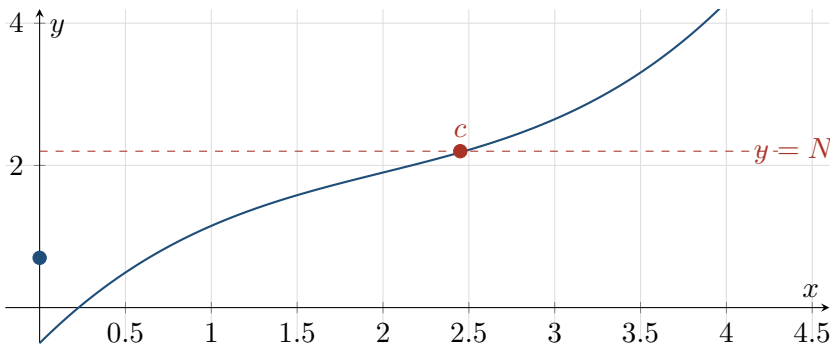


Figure 5: The intermediate value theorem: a continuous graph takes every intermediate value.

6.2 Existence of roots

A common use of the intermediate value theorem is to prove that an equation has a solution.

Example: existence of a root

Show that

$$x^3 + x - 1 = 0$$

has a solution in $(0, 1)$.
Let

$$f(x) = x^3 + x - 1.$$

This polynomial is continuous on $[0, 1]$. We compute

$$f(0) = -1, \qquad f(1) = 1.$$

Since 0 lies between -1 and 1, the intermediate value theorem gives a point $c \in (0, 1)$ such that

$$f(c) = 0.$$

Therefore the equation has at least one solution in $(0, 1)$.

6.3 Extreme value theorem

Extreme value theorem

If f is continuous on a closed bounded interval $[a, b]$, then f attains a maximum and a minimum on $[a, b]$. That is, there exist $x_{\min}, x_{\max} \in [a, b]$ such that

$$f(x_{\min}) \leq f(x) \leq f(x_{\max})$$

for all $x \in [a, b]$.

The hypotheses matter. A continuous function on an open interval need not attain its extrema. For example, $f(x) = x$ on $(0, 1)$ has no maximum and no minimum.

7 Differentiation

7.1 Average rate of change

For a function f, the average rate of change between $x = a$ and $x = a + h$ is

$$\frac{f(a + h) - f(a)}{h}, \qquad h \neq 0.$$

Geometrically, this is the slope of the secant line through the two points

$$(a, f(a)) \quad \text{and} \quad (a + h, f(a + h)).$$

7.2 Derivative as a limit

Derivative at a point

Let f be defined on an interval containing a. The derivative of f at a is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

provided this limit exists.

The quotient

$$\frac{f(a+h) - f(a)}{h}$$

is called the difference quotient. The derivative is the limiting value of this quotient as the second point approaches the first.

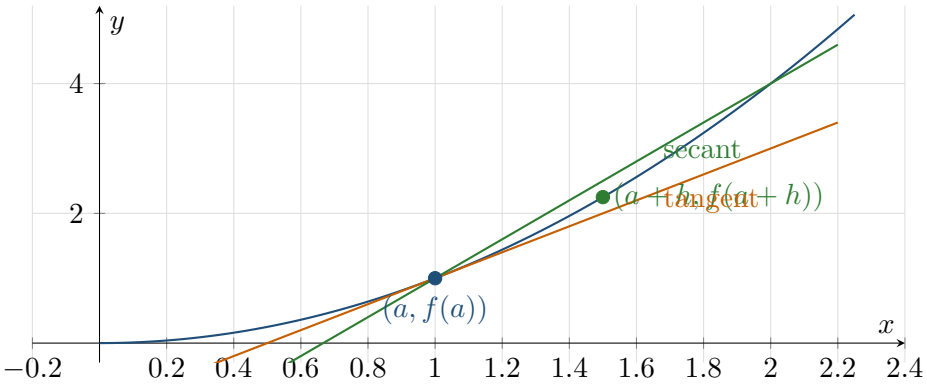


Figure 6: The derivative is obtained by letting the secant line approach the tangent line.

7.3 Equivalent form

Using $x = a + h$, one obtains the equivalent definition

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

Both forms are important. The h-form is often convenient for computations, while the x-form shows more directly that the derivative is a limit as $x \to a$.

7.4 Derivative of elementary functions from the definition

Example: derivative of x^2

Let $f(x) = x^2$. Then

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{(a+h)^2 - a^2}{h} \\ &= \lim_{h\to 0} \frac{a^2 + 2ah + h^2 - a^2}{h} \\ &= \lim_{h\to 0} (2a + h) \\ &= 2a. \end{aligned}$$

Thus

$$\frac{d}{dx} x^2 = 2x.$$

Example: derivative of $1/x$

Let $f(x) = 1/x$, with $a \neq 0$. Then

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{\frac{1}{a+h} - \frac{1}{a}}{h} \\ &= \lim_{h\to 0} \frac{\frac{a-(a+h)}{a(a+h)}}{h} \\ &= \lim_{h\to 0} \frac{-h}{a(a+h)h} \\ &= \lim_{h\to 0} \left(-\frac{1}{a(a+h)} \right) \\ &= -\frac{1}{a^2}. \end{aligned}$$

Therefore

$$\frac{d}{dx}\left(\frac{1}{x}\right) = -\frac{1}{x^2}.$$

7.5 Differentiability implies continuity

Differentiability implies continuity

If f is differentiable at a, then f is continuous at a.

Proof. We need to show that

$$\lim_{x\to a} f(x) = f(a).$$

For $x \neq a$, write

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a).$$

Taking limits as $x \to a$, the first factor tends to $f'(a)$, while the second factor tends to 0. Hence

$$\lim_{x\to a} (f(x) - f(a)) = f'(a) \cdot 0 = 0.$$

Therefore $f(x) \to f(a)$, so f is continuous at a. $\square$

The converse is false: a function can be continuous but not differentiable.

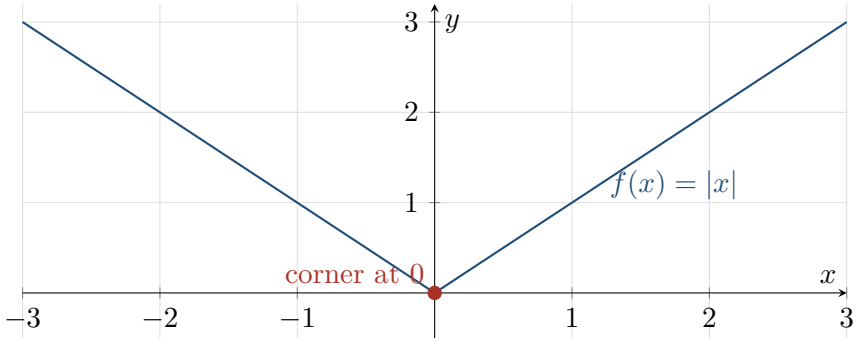


Figure 7: The function $f(x) = |x|$ is continuous at 0, but it is not differentiable at 0.

For $f(x) = |x|$, the right derivative at 0 is 1, while the left derivative is -1. Since they are not equal, the derivative at 0 does not exist.

7.6 Tangent line and local linear approximation

If f is differentiable at a, then near a the function is well approximated by the tangent line

$$L(x) = f(a) + f'(a)(x - a).$$

This is called the linearization of f at a.

More precisely,

$$f(x) = f(a) + f'(a)(x - a) + r(x),$$

where the remainder satisfies

$$\lim_{x \to a} \frac{r(x)}{x - a} = 0.$$

Thus the error is small compared with $|x - a|$.

8 Rules of differentiation

8.1 Linearity

Linearity rule

If f and g are differentiable at x, and $\alpha, \beta \in \mathbb{R}$, then

$$(\alpha f + \beta g)'(x) = \alpha f'(x) + \beta g'(x).$$

8.2 Product rule

Product rule

If f and g are differentiable at x, then

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x).$$

Idea of proof. Add and subtract the same term:

$$\begin{aligned}\frac{f(x+h)g(x+h)-f(x)g(x)}{h} &= \frac{f(x+h)g(x+h)-f(x)g(x+h)}{h} \\ &\quad + \frac{f(x)g(x+h)-f(x)g(x)}{h} \\ &= g(x+h)\frac{f(x+h)-f(x)}{h} + f(x)\frac{g(x+h)-g(x)}{h}.\end{aligned}$$

Letting $h \to 0$ gives the result, because differentiability implies continuity.

8.3 Quotient rule

Quotient rule

If f and g are differentiable at x, and $g(x) \neq 0$, then

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{g(x)^2}.$$

8.4 Chain rule

Chain rule

If g is differentiable at x, and f is differentiable at $g(x)$, then

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

The chain rule is the rule for differentiating a composition. It is one of the most important computational tools in calculus.

Example: chain rule

Let

$$y = \sin(x^2 + 1).$$

Set $u = x^2 + 1$. Then $y = \sin u$, so

$$\frac{dy}{dx} = \cos(u)\frac{du}{dx} = \cos(x^2+1) \cdot 2x.$$

Therefore

$$y' = 2x\cos(x^2+1).$$

8.5 Power rule

For integer powers, the power rule states

$$\frac{d}{dx}x^n = nx^{n-1}.$$

For positive integers, this follows from algebra and the definition of derivative. With more theory, it extends to rational and real exponents on appropriate domains.

Example: polynomial derivative

Let

$$f(x) = 4x^5 - 3x^2 + 7x - 1.$$

Then

$$f'(x) = 20x^4 - 6x + 7.$$

9 Derivatives of standard functions

Function	Derivative
c	0
x^n	nx^{n-1}
e^x	e^x
a^x, $a > 0$	$a^x \ln a$
$\ln x$, $x > 0$	$1/x$
$\sin x$	$\cos x$
$\cos x$	$-\sin x$
$\tan x$	$\sec^2 x$

These formulas should not be treated as isolated facts. Each one is connected to a defining limit, a structural rule, or both.

10 Higher derivatives and shape of a graph

10.1 Second derivative

If f' is differentiable, then the derivative of f' is the second derivative:

$$f''(x) = \frac{d}{dx} f'(x).$$

The first derivative describes the rate of change of f. The second derivative describes the rate of change of the rate of change.

10.2 Monotonicity

Derivative sign and monotonicity

Let f be differentiable on an interval I.

- If $f'(x) > 0$ for all $x \in I$, then f is strictly increasing on I.
- If $f'(x) < 0$ for all $x \in I$, then f is strictly decreasing on I.

10.3 Concavity

A function is concave up on an interval if its graph bends upward. A practical criterion is:

$$f''(x) > 0 \quad \Longrightarrow \quad \text{concave up},$$

while

$$f''(x) < 0 \quad \Longrightarrow \quad \text{concave down}.$$

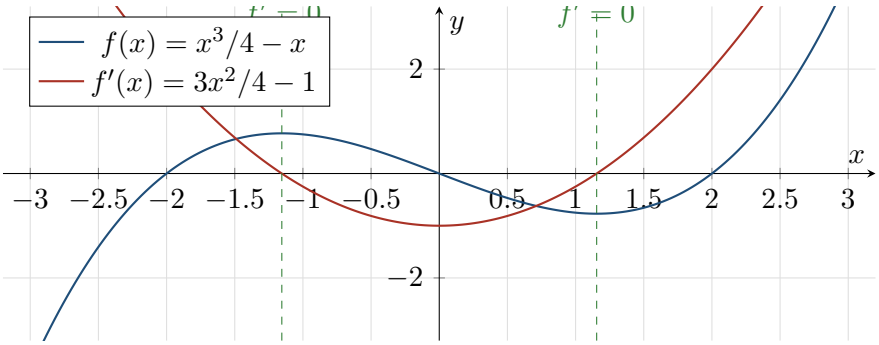


Figure 8: Critical points occur where $f'(x) = 0$ or where the derivative does not exist.

10.4 Critical points

Critical point

A point c in the domain of f is called a critical point if either

$$f'(c) = 0$$

or $f'(c)$ does not exist.

Critical points are candidates for local extrema, but not every critical point is an extremum.

11 Rolle's theorem and the mean value theorem

11.1 Rolle's theorem

Rolle's theorem

Suppose that f is continuous on $[a, b]$, differentiable on (a, b), and

$$f(a) = f(b).$$

Then there exists $c \in (a, b)$ such that

$$f'(c) = 0.$$

Geometrically, if a smooth curve starts and ends at the same height, then somewhere in between it has a horizontal tangent.

11.2 Mean value theorem

Mean value theorem

Suppose that f is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

The mean value theorem says that somewhere on the interval, the instantaneous rate of change equals the average rate of change over the whole interval.

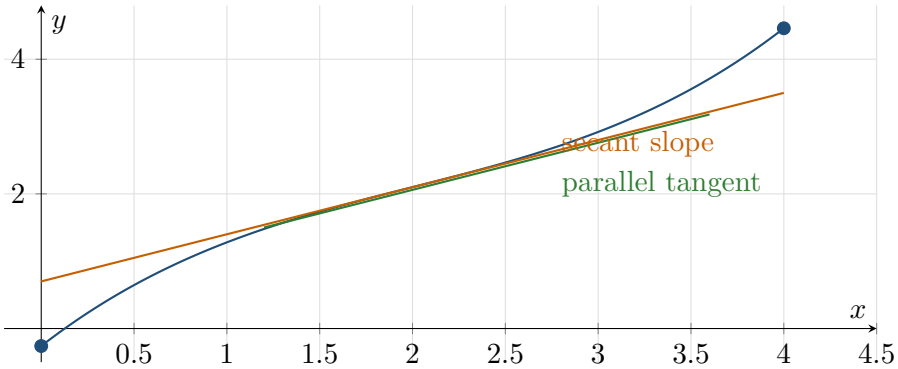


Figure 9: The mean value theorem: at some point, the tangent is parallel to the secant line.

11.3 Consequences of the mean value theorem

Zero derivative implies constant function

If $f'(x) = 0$ for all $x \in (a, b)$, then f is constant on (a, b).

Reason. For any two points $x_1 < x_2$, the mean value theorem gives

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$$

for some $c \in (x_1, x_2)$. Thus $f(x_2) = f(x_1)$.

12 Taylor's first approximation

Differentiability means that near a, the function behaves like its tangent line. This gives the first-order Taylor approximation:

$$f(x) \approx f(a) + f'(a)(x - a).$$

Example: approximating $\sqrt{4.1}$

Let $f(x) = \sqrt{x}$, and choose $a = 4$. Then

$$f(4) = 2, \qquad f'(x) = \frac{1}{2\sqrt{x}}, \qquad f'(4) = \frac{1}{4}.$$

Thus

$$\sqrt{x} \approx 2 + \frac{1}{4}(x-4).$$

For $x = 4.1$,

$$\sqrt{4.1} \approx 2 + \frac{1}{4}(0.1) = 2.025.$$

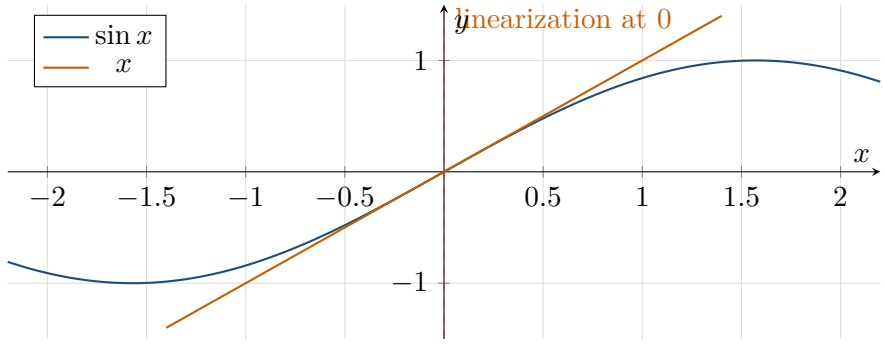


Figure 10: Near 0, $\sin x$ is well approximated by its tangent line $y = x$.

13 Non-differentiability: what can go wrong?

A function may fail to be differentiable at a point for several reasons.

13.1 Corners

The function $f(x) = |x|$ has a sharp corner at 0. The left and right slopes disagree.

13.2 Cusps

The function $f(x) = |x|^{2/3}$ has a cusp at 0. The tangent becomes vertical in a singular way.

13.3 Vertical tangents

The function $f(x) = \sqrt[3]{x}$ has a vertical tangent at 0. It is continuous, but its derivative is unbounded near 0.

13.4 Discontinuities

If a function is not continuous at a point, it cannot be differentiable there.

Hierarchy of conditions

The logical implications are:

$$\text{differentiable at } a \quad \Longrightarrow \quad \text{continuous at } a \quad \Longrightarrow \quad \lim_{x \to a} f(x) = f(a).$$

The reverse implications are false in general.

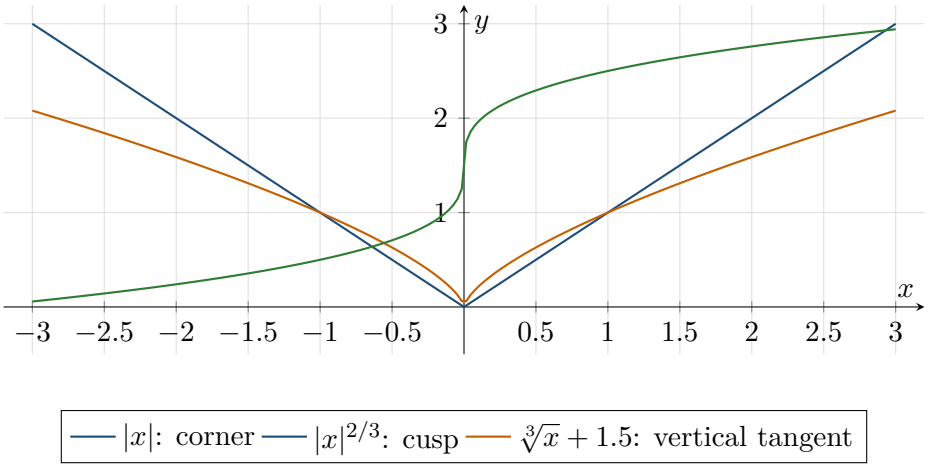


Figure 11: Several ways differentiability can fail while continuity may still hold.

14 Worked examples

Example 1: limit with rationalization

Compute

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x}.$$

Multiply by the conjugate:

$$\begin{aligned}\frac{\sqrt{1+x}-1}{x} &= \frac{\sqrt{1+x}-1}{x}\cdot\frac{\sqrt{1+x}+1}{\sqrt{1+x}+1}\\ &= \frac{(1+x)-1}{x(\sqrt{1+x}+1)}\\ &= \frac{1}{\sqrt{1+x}+1}.\end{aligned}$$

Therefore

$$\lim_{x\to 0} \frac{\sqrt{1+x}-1}{x} = \frac{1}{2}.$$

Example 2: continuity of a piecewise function

Find c such that

$$f(x) = \begin{cases} 2x+1, & x<1,\\ cx^2, & x\geq 1\end{cases}$$

is continuous at $x=1$.
The left limit is

$$\lim_{x\to 1^-}(2x+1) = 3.$$

The value at 1 is

$$f(1) = c.$$

For continuity, we need $c=3$.

Example 3: differentiability of a piecewise function

Find a, b such that

$$f(x) = \begin{cases} x^2, & x < 1, \\ ax + b, & x \geq 1 \end{cases}$$

is differentiable at $x = 1$.
Differentiability implies continuity, so first require

$$\lim_{x \to 1^-} x^2 = f(1) = a + b.$$

Thus

$$a + b = 1.$$

Now match derivatives. The derivative from the left is

$$\lim_{x \to 1^-} 2x = 2.$$

The derivative from the right is a. Hence

$$a = 2.$$

Then $a + b = 1$ gives $b = -1$. Therefore

$$a = 2, \qquad b = -1.$$

Example 4: using the mean value theorem

Show that

$$|\sin x - \sin y| \leq |x - y|$$

for all real x, y.
Assume $x \neq y$. By the mean value theorem applied to $f(t) = \sin t$ on the interval between x and y, there exists c between them such that

$$\frac{\sin x - \sin y}{x - y} = \cos c.$$

Taking absolute values,

$$|\sin x - \sin y| = |\cos c| \; |x - y| \leq |x - y| \, .$$

If $x = y$, the inequality is obvious.

15 Proof templates

15.1 Template for proving a finite limit

To prove $\lim_{x \to a} f(x) = L$:

1. Start with $|f(x) - L|$.
2. Rewrite or estimate it in terms of $|x - a|$.
3. Control any additional factors by restricting x to a small neighborhood of a.

4. Choose δ as the minimum of all needed restrictions.
5. Write the formal implication.

15.2 Template for proving continuity

To prove that f is continuous at a:

1. Verify that $f(a)$ is defined.
2. Show that $\lim_{x\to a} f(x)$ exists.
3. Show that the limit equals $f(a)$.

Alternatively, use known theorems: polynomials, rational functions on their domains, compositions, sums, products, and quotients of continuous functions.

15.3 Template for proving differentiability

To prove differentiability at a:

1. Write the difference quotient:
$$\frac{f(a+h)-f(a)}{h}.$$
2. Simplify the expression for $h \neq 0$.
3. Take the limit as $h \to 0$.
4. If the limit exists, it is $f'(a)$.

For piecewise functions, compute one-sided derivatives and compare them.

16 Exercises

A. Limits

1. Prove using the epsilon-delta definition that
$$\lim_{x\to 3}(2x+4) = 10.$$
2. Compute
$$\lim_{x\to 1}\frac{x^2-1}{x-1}.$$
3. Compute
$$\lim_{x\to 0}\frac{\sin(5x)}{x}.$$
4. Determine whether the limit exists:
$$\lim_{x\to 0}\frac{|x|}{x}.$$
5. Use the squeeze theorem to show that
$$\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0.$$

B. Continuity

1. Determine all points of discontinuity of
$$f(x) = \frac{x^2 - 4}{x - 2}.$$
2. Find a such that
$$f(x) = \begin{cases} ax + 1, & x < 2, \\ x^2, & x \geq 2 \end{cases}$$
is continuous at $x = 2$.
3. Prove that the equation
$$x^5 + x - 3 = 0$$
has at least one solution in $(1, 2)$.
4. Give an example of a function that has a limit at 0 but is not continuous at 0.

C. Differentiation

1. Compute the derivative of $f(x) = x^3$ from the definition.
2. Determine whether $f(x) = |x - 2|$ is differentiable at $x = 2$.
3. Find the tangent line to
$$f(x) = x^2 + 1$$
at $x = 1$.
4. Differentiate
$$f(x) = \frac{x^2 + 1}{x - 1}.$$
5. Find intervals of increase and decrease for
$$f(x) = x^3 - 3x.$$
6. Use the mean value theorem to prove that
$$|\cos x - \cos y| \leq |x - y|$$
for all real x, y.

17 Selected solutions

A. Limits

A1. We want
$$|(2x + 4) - 10| = 2\,|x - 3| < \varepsilon.$$
Choose $\delta = \varepsilon/2$. Then $0 < |x - 3| < \delta$ implies the required inequality.

A2.
$$\frac{x^2 - 1}{x - 1} = x + 1$$

for $x \neq 1$, so the limit is 2.

A3.

$$\frac{\sin(5x)}{x} = 5 \cdot \frac{\sin(5x)}{5x} \to 5.$$

A4. For $x > 0$, $|x|/x = 1$. For $x < 0$, $|x|/x = -1$. The one-sided limits are different, so the two-sided limit does not exist.

A5. Since $-1 \leq \cos(1/x) \leq 1$,

$$-|x| \leq x\cos(1/x) \leq |x|.$$

Both bounds tend to 0, so the limit is 0.

B. Continuity

B1. The formula is undefined at $x = 2$. For $x \neq 2$, it simplifies to $x + 2$, so the limit at 2 is 4. Thus there is a removable discontinuity at 2.

B2. The left limit at 2 is $2a + 1$, while the value from the second branch is 4. Continuity requires $2a + 1 = 4$, hence $a = 3/2$.

B3. Let $f(x) = x^5 + x - 3$. Then $f(1) = -1$, $f(2) = 31$. By the intermediate value theorem, there is a root in $(1, 2)$.

B4. For example,

$$f(x) = \begin{cases} 0, & x \neq 0, \\ 1, & x = 0. \end{cases}$$

Then $\lim_{x\to 0} f(x) = 0$, but $f(0) = 1$, so f is not continuous at 0.

C. Differentiation

C1.

$$\begin{aligned} f'(a) &= \lim_{h\to 0} \frac{(a+h)^3 - a^3}{h} \\ &= \lim_{h\to 0} \frac{3a^2h + 3ah^2 + h^3}{h} \\ &= \lim_{h\to 0} (3a^2 + 3ah + h^2) = 3a^2. \end{aligned}$$

Thus $(x^3)' = 3x^2$.

C2. At $x = 2$, the left derivative is -1 and the right derivative is 1. Therefore the derivative does not exist at 2.

C3. Since $f(1) = 2$ and $f'(x) = 2x$, we have $f'(1) = 2$. The tangent line is

$$y - 2 = 2(x - 1),$$

so $y = 2x$.

C4. By the quotient rule,

$$f'(x) = \frac{2x(x-1) - (x^2+1)}{(x-1)^2} = \frac{x^2 - 2x - 1}{(x-1)^2}.$$

C5. For $f(x) = x^3 - 3x$,

$$f'(x) = 3x^2 - 3 = 3(x-1)(x+1).$$

Thus $f' > 0$ on $(-\infty, -1) \cup (1, \infty)$, and $f' < 0$ on $(-1, 1)$. The function is increasing on $(-\infty, -1)$ and $(1, \infty)$, decreasing on $(-1, 1)$.

C6. Apply the mean value theorem to $f(t) = \cos t$. There exists c between x and y such that

$$\frac{\cos x - \cos y}{x - y} = -\sin c.$$

Hence

$$|\cos x - \cos y| = |\sin c|\,|x - y| \le |x - y|.$$

18 Compact formula sheet

Limits and continuity

$$\lim_{x\to a} f(x) = L \quad \Longleftrightarrow \quad \forall \varepsilon > 0\ \exists \delta > 0 :\ 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

$$f \text{ continuous at } a \quad \Longleftrightarrow \quad \lim_{x\to a} f(x) = f(a).$$

$$\lim_{x\to 0} \frac{\sin x}{x} = 1, \qquad \lim_{x\to 0} \frac{e^x - 1}{x} = 1, \qquad \lim_{x\to 0} \frac{\ln(1+x)}{x} = 1.$$

Differentiation

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h} = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}.$$

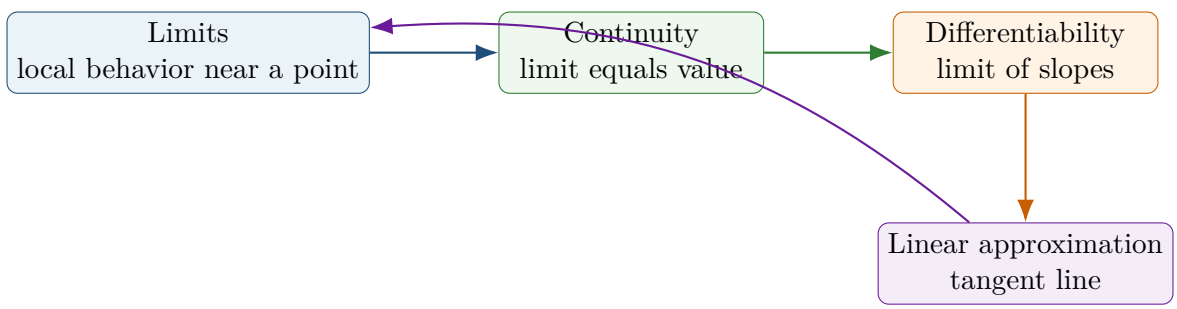
$$(f+g)' = f' + g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}.$$

$$(f \circ g)'(x) = f'(g(x))g'(x).$$

$$\frac{d}{dx} x^n = n x^{n-1}, \qquad \frac{d}{dx} e^x = e^x, \qquad \frac{d}{dx} \ln x = \frac{1}{x}.$$

$$\frac{d}{dx} \sin x = \cos x, \qquad \frac{d}{dx} \cos x = -\sin x.$$

19 Final conceptual map



The conceptual direction is:

$$\text{local approximation of values} \quad \to \quad \text{no jumps} \quad \to \quad \text{local approximation by lines.}$$

This is the foundation for later topics: optimization, Taylor series, integration, differential equations, multivariable calculus, and mathematical modeling.